15 $x = 0.4\dot{3}\dot{6}$

Prove algebraically that x can be written as $\frac{24}{55}$

$x = 0.4363636...$ The repeating block is 36.

$10x = 4.363636...$

1 mark
Align the same repeating decimal tail

$1000x = 436.363636...$

Subtract: $1000x - 10x = 436.3636... - 4.3636...$

$990x = 432$

1 mark
The repeating tails cancel

$x = \frac{432}{990} = \frac{24}{55}$

1 mark
Divide by 990, then simplify by 18

(Total for Question 15 is 3 marks)